```
r1.upper.middle, step 2 (2026-11-02)

 chain     seed iterations.saved acceptance.rate final.log.likelihood best.log.likelihood errored.proposals
     1 20263016             5000           0.276             -38371.2            -38367.3                 0
     2 20263017             5000           0.277             -38377.9            -38365.1                 0
     3 20263018             5000           0.276             -38400.3            -38379.5                 0
     4 20263019             5000           0.275             -38390.3            -38375.4                 0
                                                file
 mcmc_r1.upper.middle_step2_chain1_2026-10-30.Rdata
 mcmc_r1.upper.middle_step2_chain2_2026-10-30.Rdata
 mcmc_r1.upper.middle_step2_chain3_2026-10-30.Rdata
 mcmc_r1.upper.middle_step2_chain4_2026-10-30.Rdata

Worst R-hats (> 1.1 suggests the chains haven't converged):
    age.65.to.79.mortality.slope.multiplier.female                          female.to.male.multiplier
                                             2.303                                              2.218
                                           trate.1              age.30.to.39.transmission.multiplier.0
                                             1.882                                              1.820
                                           trate.3                      hiv.specific.mortality.rates.1
                                             1.815                                              1.746
            age.20.to.29.transmission.multiplier.0 age.65.to.79.mortality.intercept.multiplier.female
                                             1.693                                              1.630
                                           trate.0              age.20.to.29.transmission.multiplier.3
                                             1.614                                              1.495
          age.15.to.24.hiv.mortality.multiplier.1             age.25.to.49.hiv.mortality.multiplier.1
                                             1.493                                              1.481
        age.50.and.over.transmission.multiplier.3              age.20.to.29.transmission.multiplier.1
                                             1.477                                              1.404
            age.0.to.14.hiv.mortality.multiplier.1                    male.hiv.mortality.multiplier.0
                                             1.388                                              1.370
                                           trate.4              age.30.to.39.transmission.multiplier.1
                                             1.356                                              1.322
            age.15.to.19.transmission.multiplier.0              age.15.to.19.transmission.multiplier.3
                                             1.315                                              1.313
  age.65.to.79.mortality.intercept.multiplier.male              age.0.to.14.hiv.mortality.multiplier.2
                                             1.261                                              1.256
         over.80.mortality.slope.multiplier.female             over.80.mortality.slope.multiplier.male
                                             1.255                                              1.252
                                           trate.2
                                             1.228

Merged simset:
        location simulations total.weight                                                    file
 r1.upper.middle        1000         1000 simset_r1.upper.middle_step2_chains1234_2026-10-30.Rdata

Fit check (merged simset vs every calibration target, 2000-2025): 22 of 33 targets flagged
     data.type         age    sex years.with.data pct.years.in.sim.95 median.sim.vs.data last.year last.sim.vs.data
 hiv.mortality 50 and over   both              26                  12              x2.54      2025            x2.45
 hiv.mortality       15-24   both              26                   4              x1.99      2025            x2.08
 hiv.mortality       10-19   both              26                   0              x1.93      2025            x1.94
 hiv.mortality        0-14   both              26                   8              x1.68      2025            x1.60
     awareness        0-14   both              11                   0          -20.7 pts      2025        -29.3 pts
 hiv.mortality       25-49   both              26                   8              x0.73      2025            x0.89
     incidence       10-19   both              26                   8              x0.75      2025            x0.79
   suppression        0-14   both              11                  27           +7.6 pts      2025        +10.6 pts
    prevalence       15-24   both              26                  15              x1.14      2025            x1.21
```

r1.upper.middle step 2: likelihood

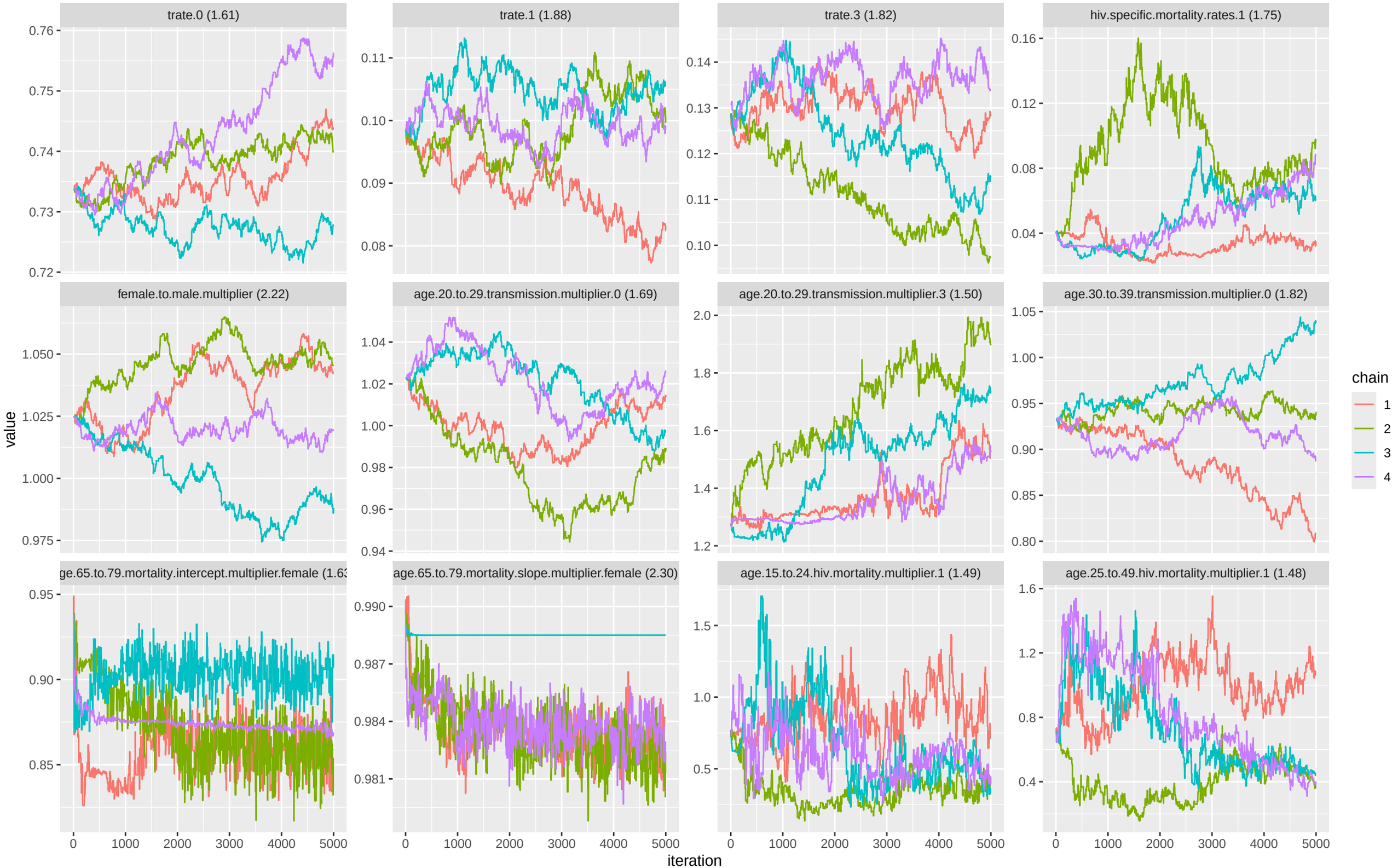
0
−10000
−20000
−30000
−40000
lue

0
25000
50000
75000
100000
iteration

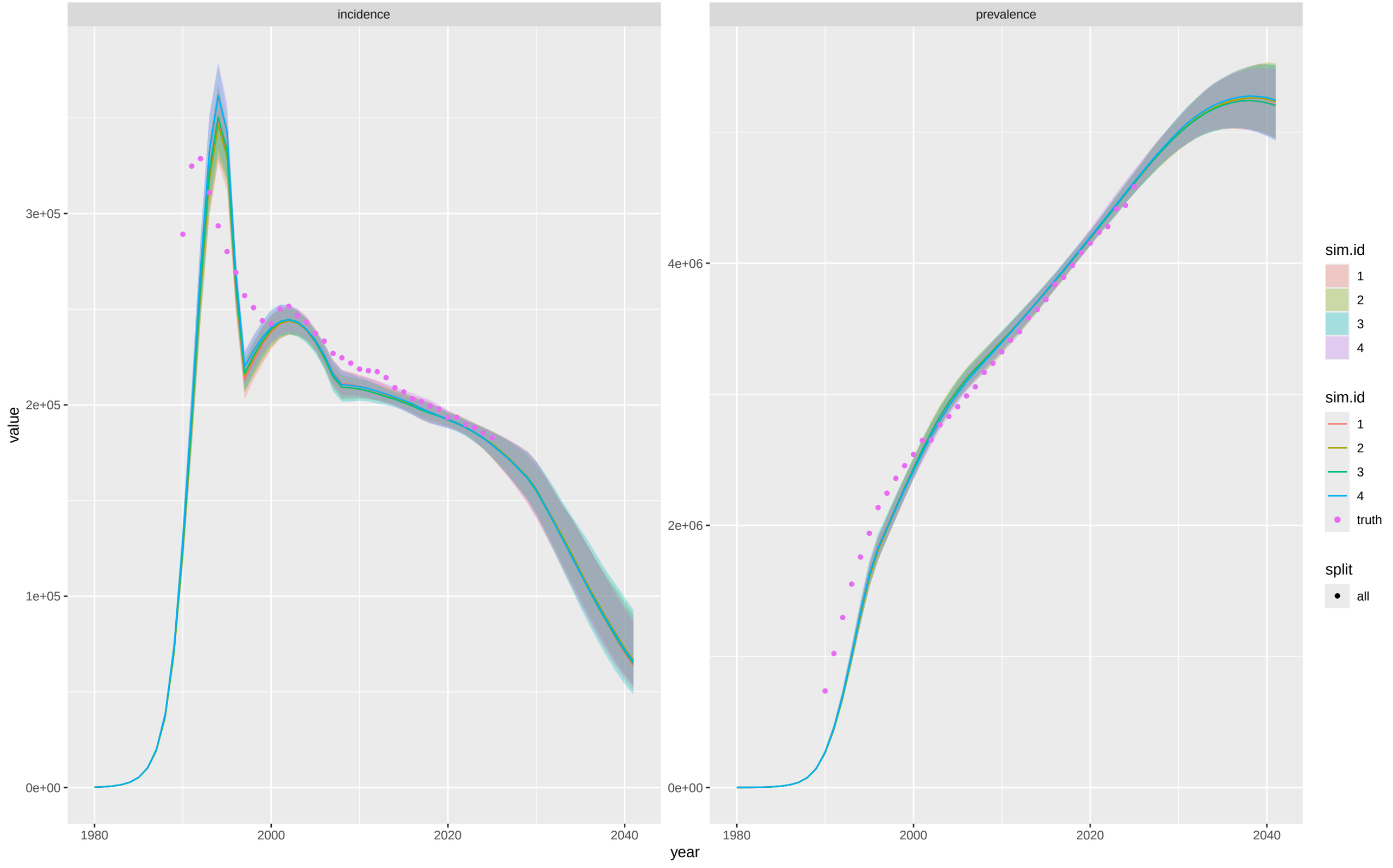
chain
1
2
3
4

type
Log Likelihood
Log Prior
Log Prior x Likelihood

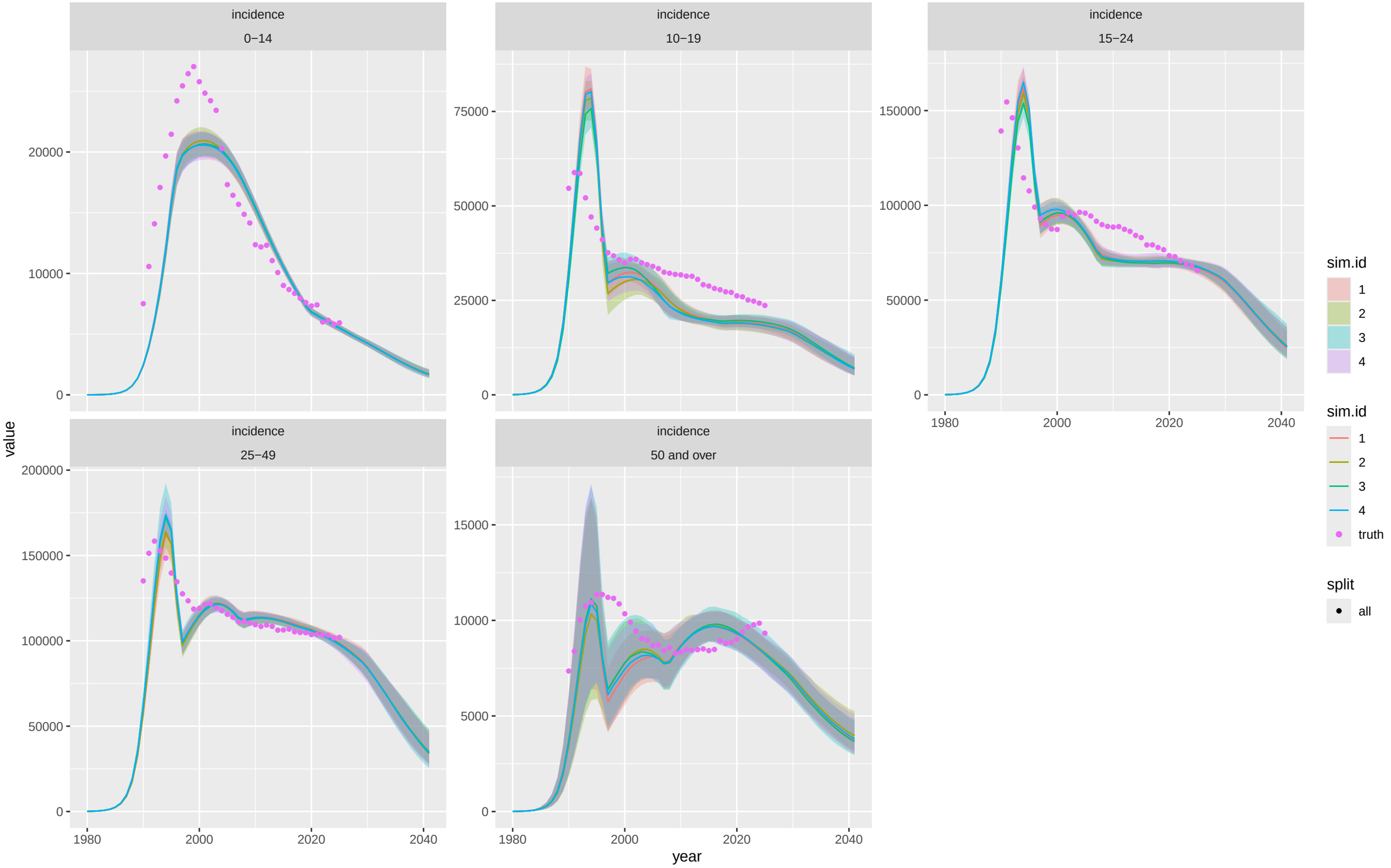
r1.upper.middle step 2: trace



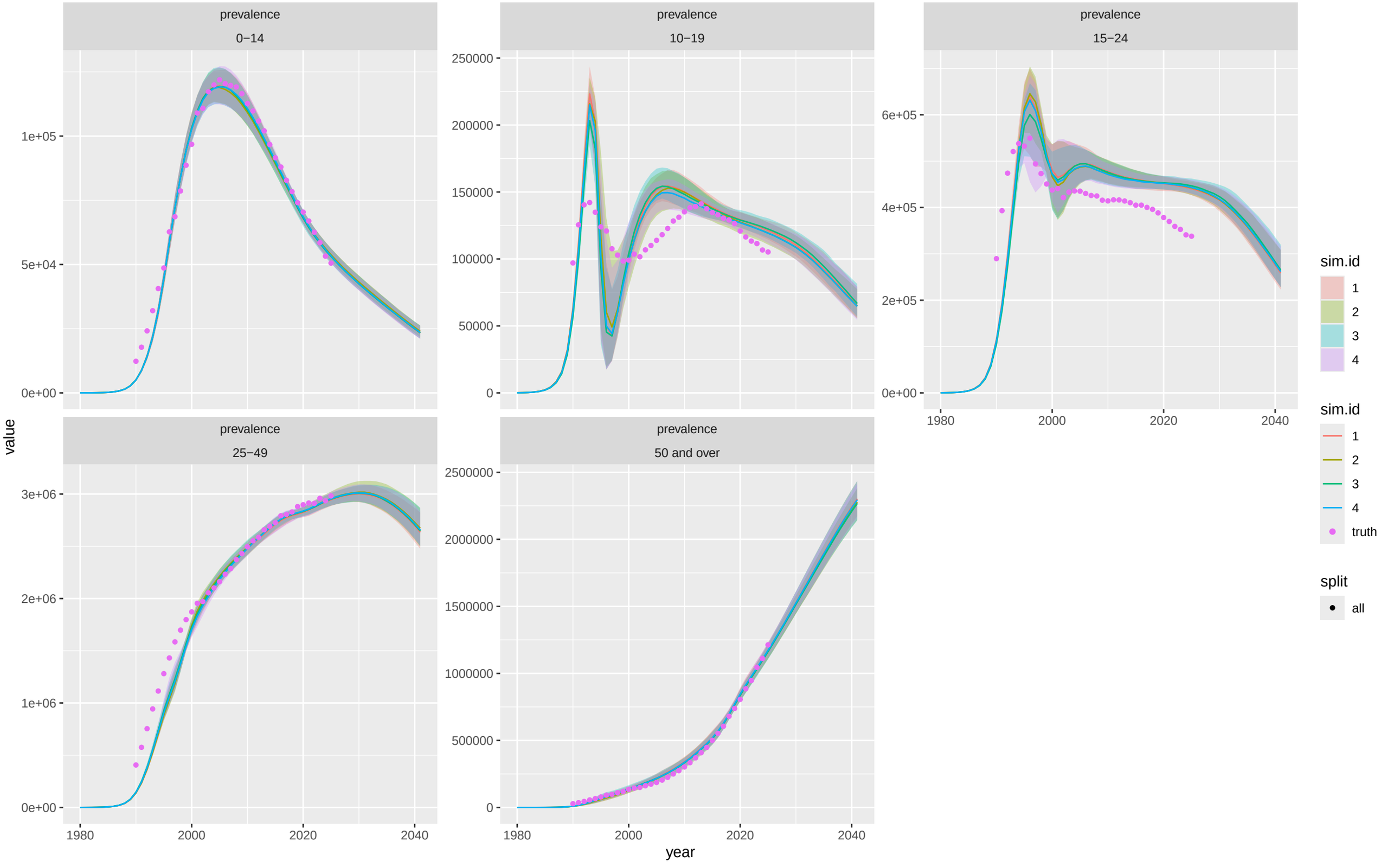
r1.upper.middle step 2: fit (one color per chain)



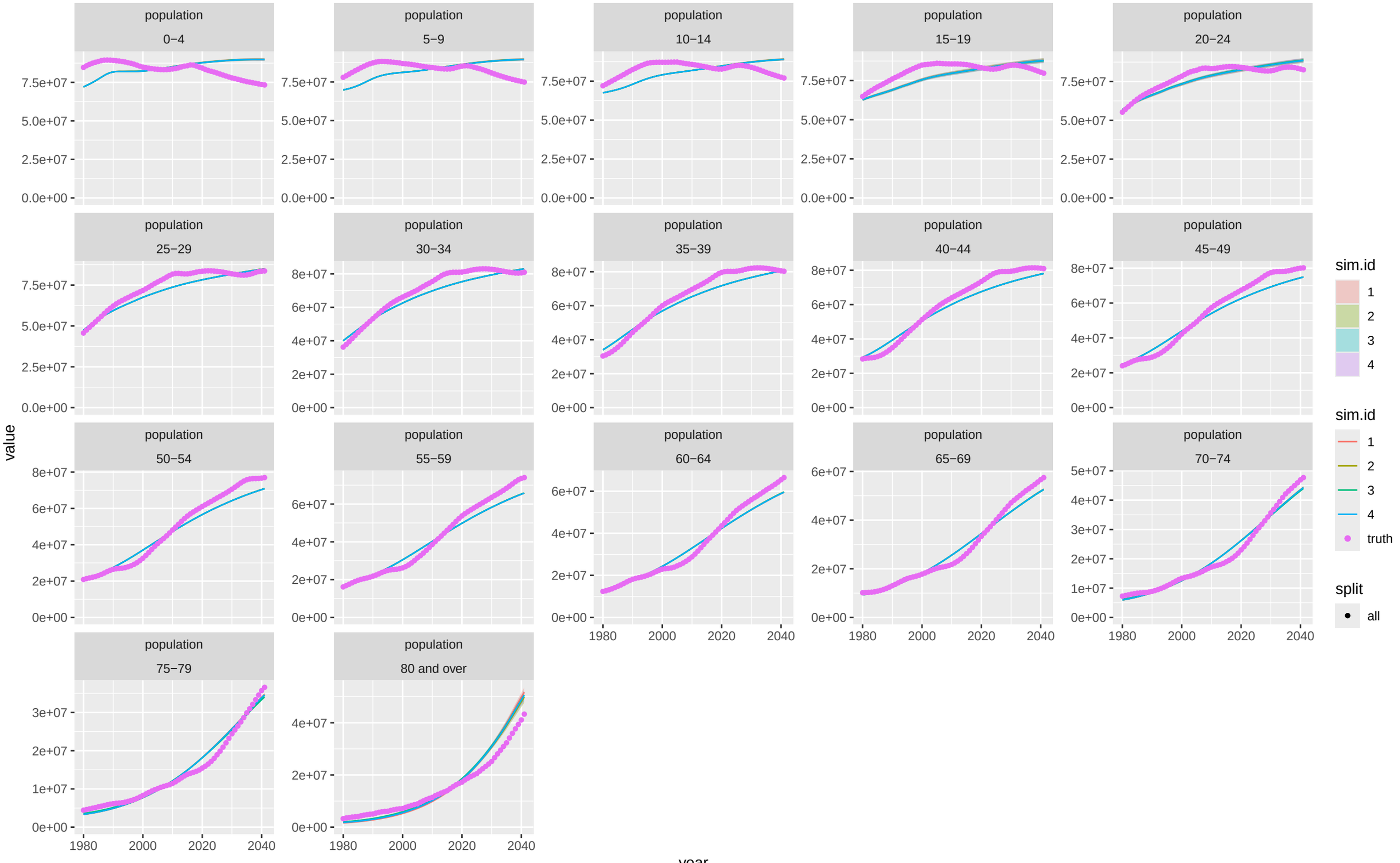
r1.upper.middle step 2: incidence.by.age (one color per chain)



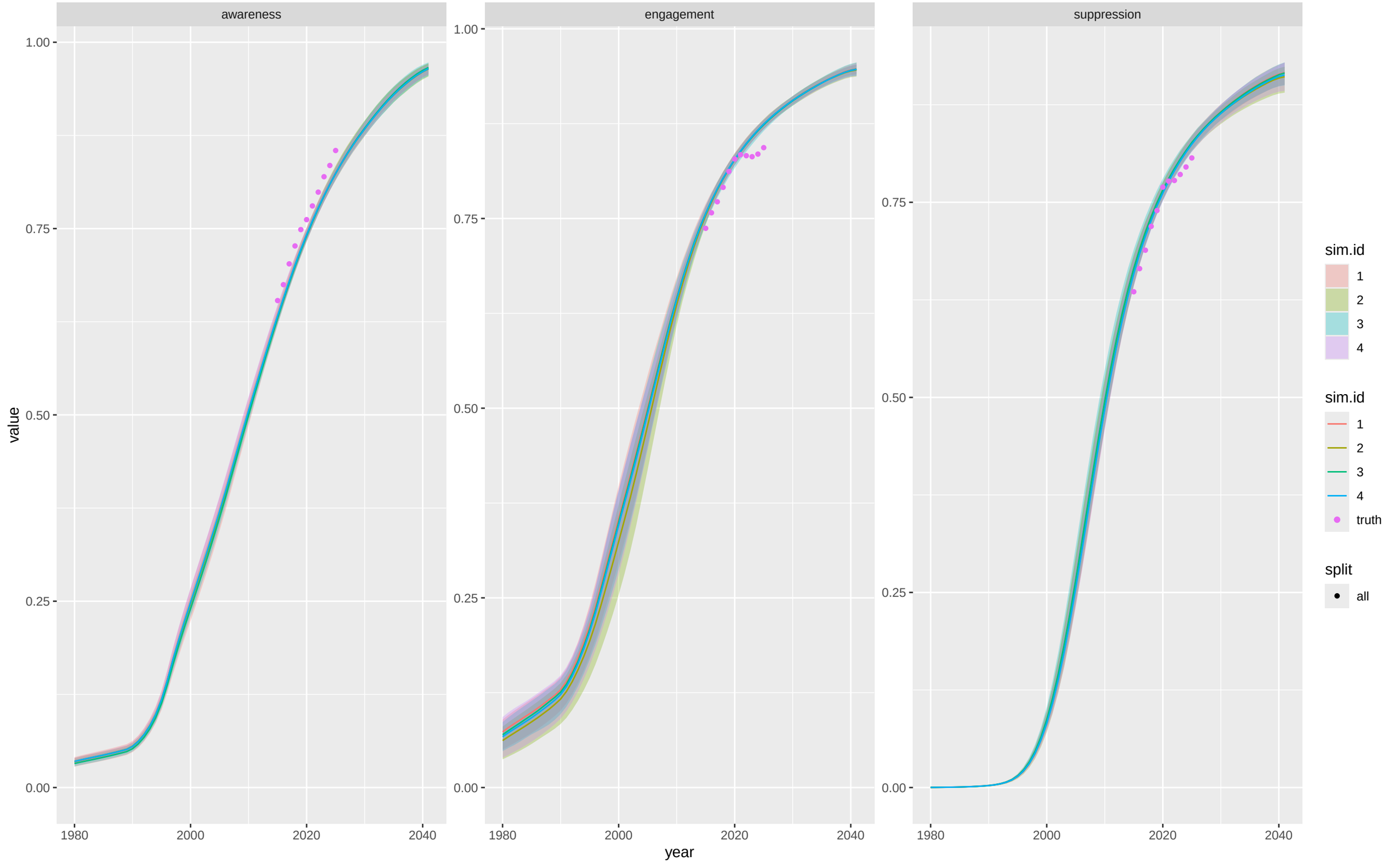
r1.upper.middle step 2: prevalence.by.age (one color per chain)



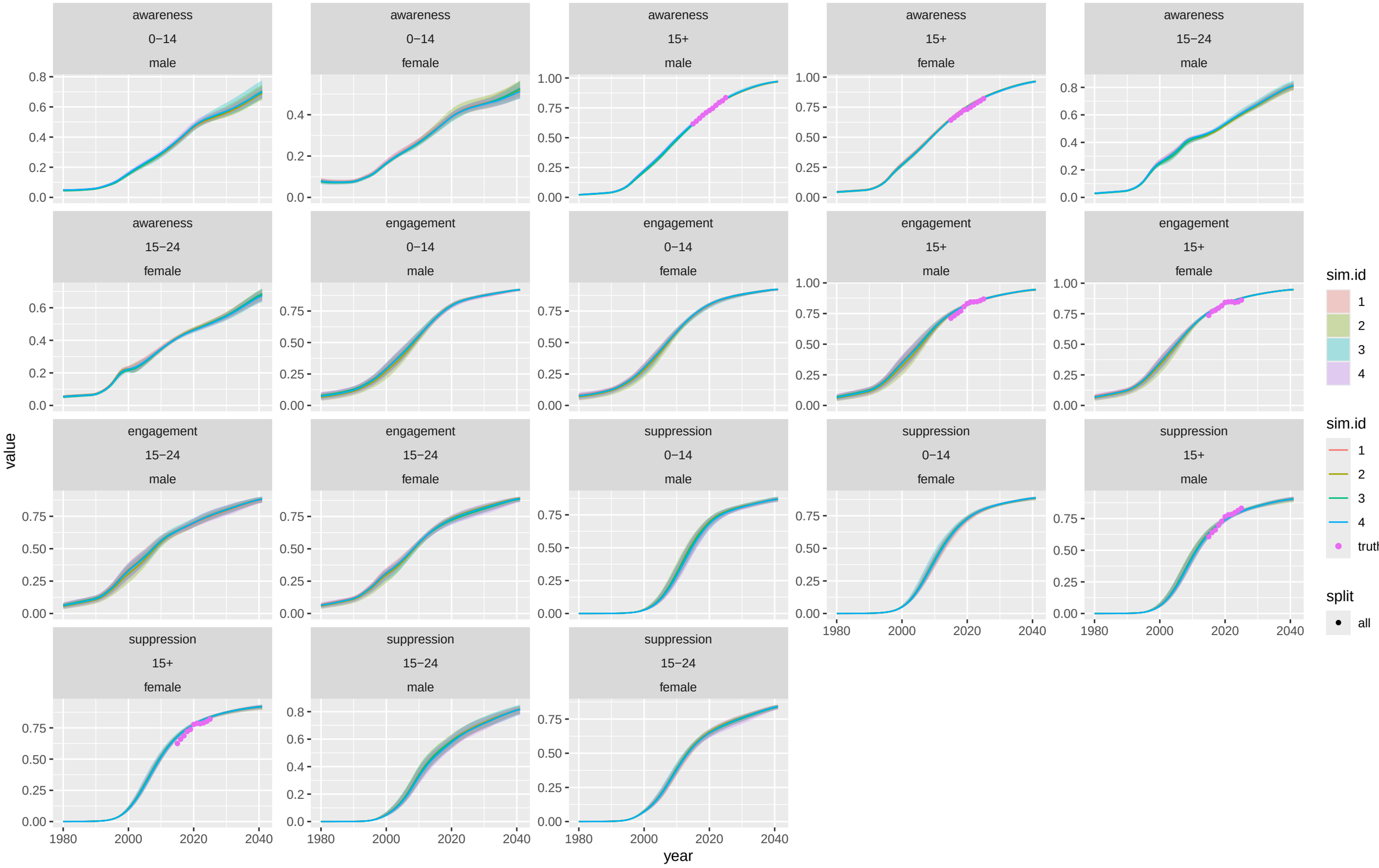
r1.upper.middle step 2: population.by.age (one color per chain)



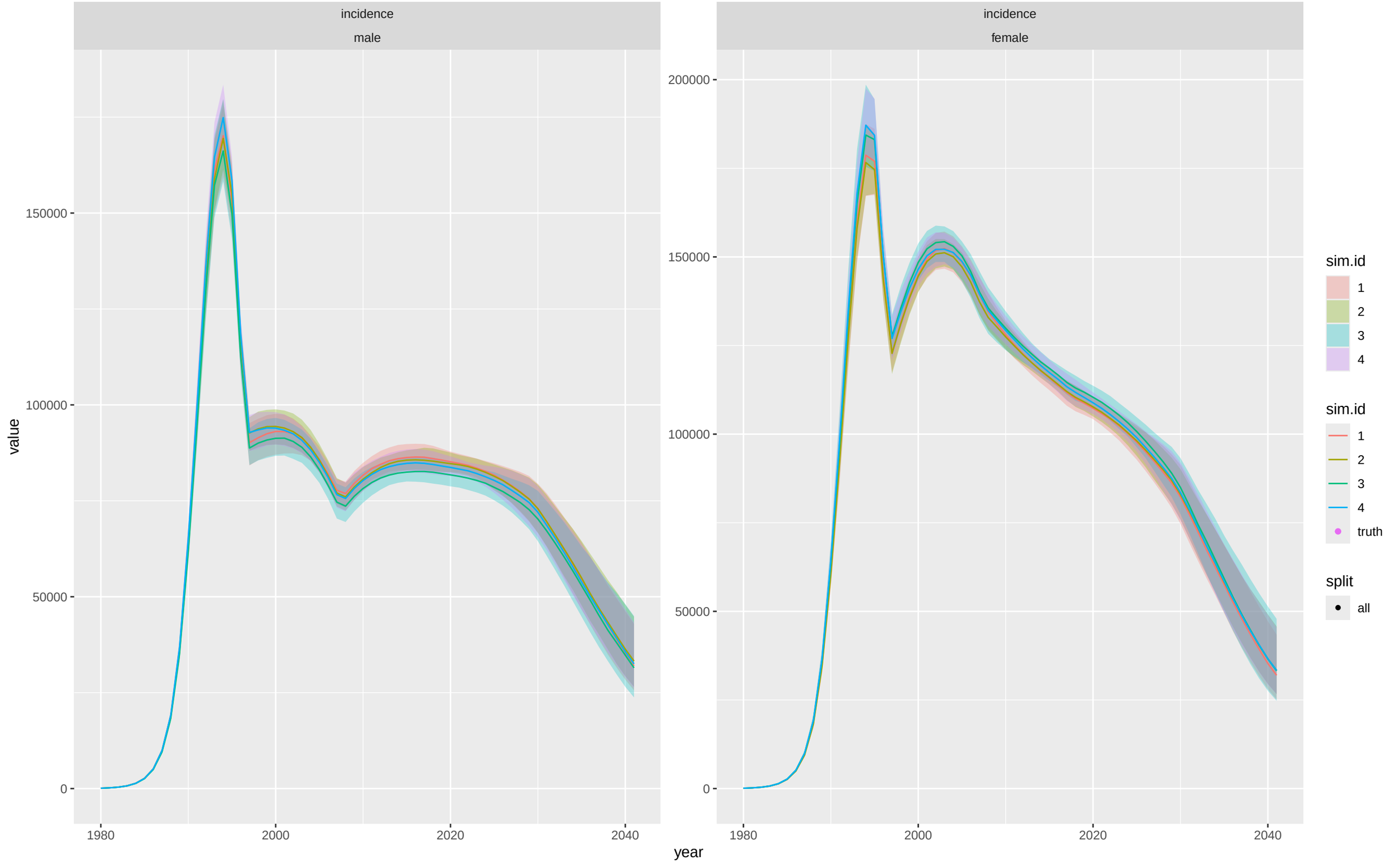
r1.upper.middle step 2: cascade (one color per chain)



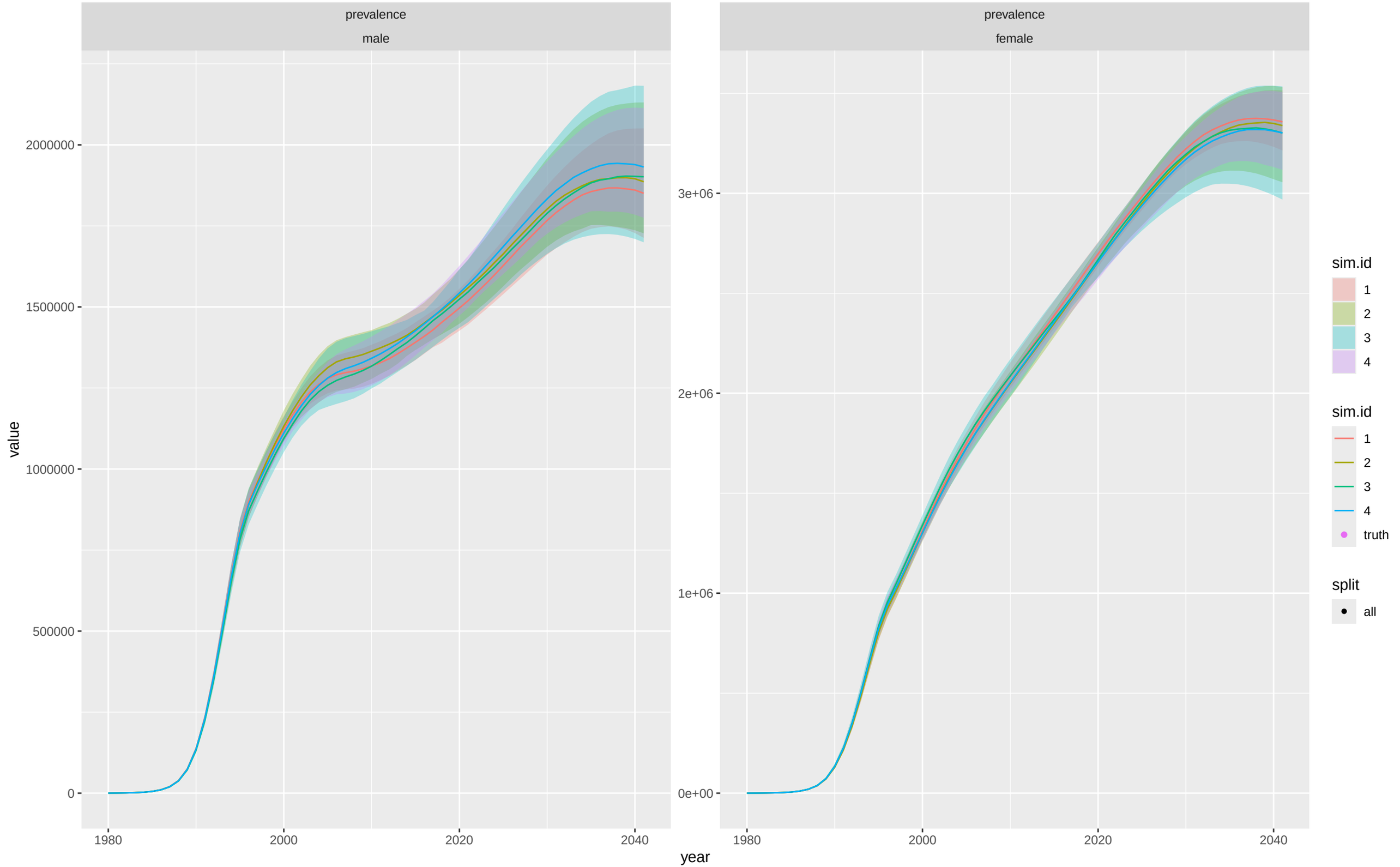
r1.upper.middle step 2: cascade.by.age.sex (one color per chain)



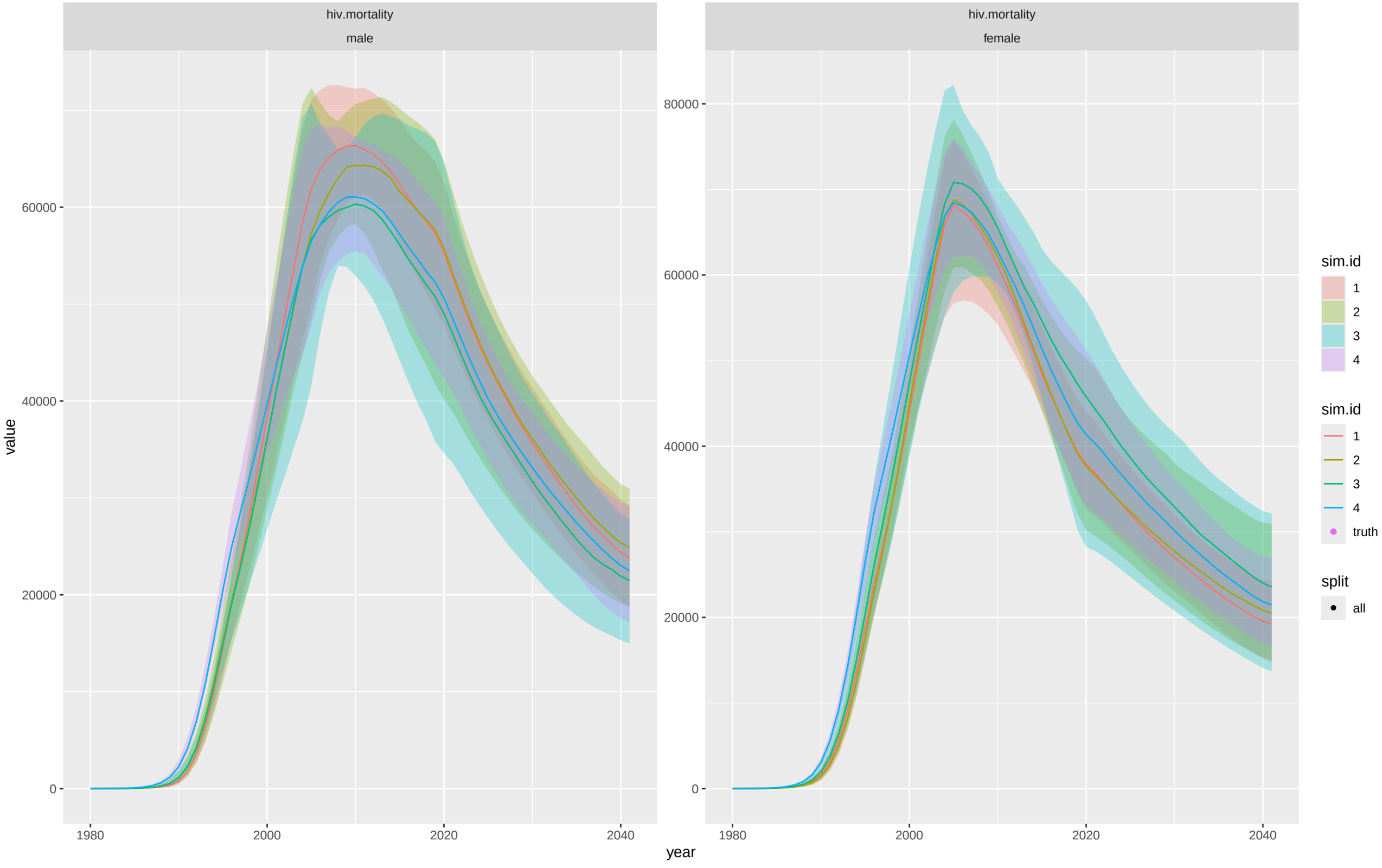
r1.upper.middle step 2: incidence.by.sex (one color per chain)



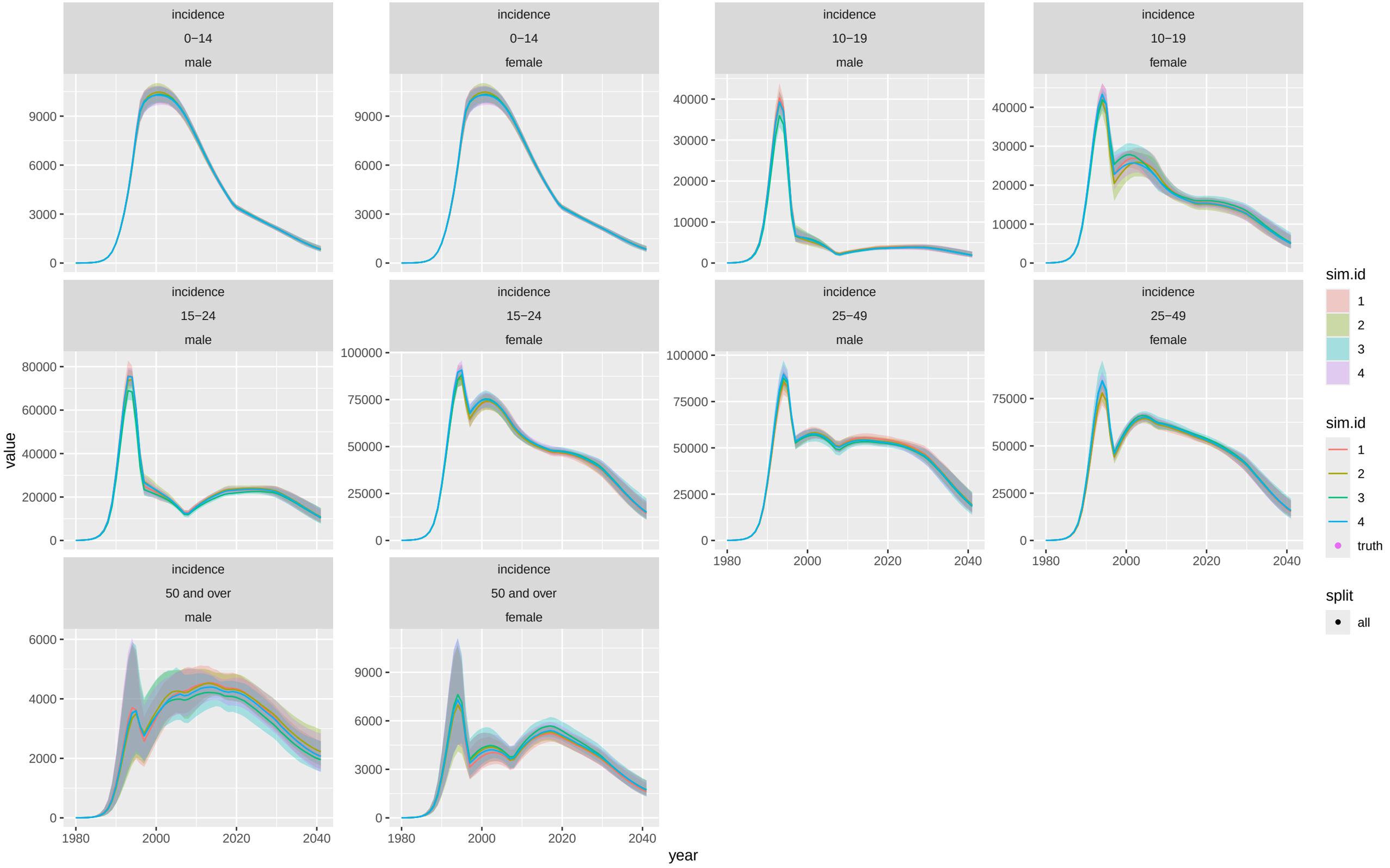
r1.upper.middle step 2: prevalence.by.sex (one color per chain)



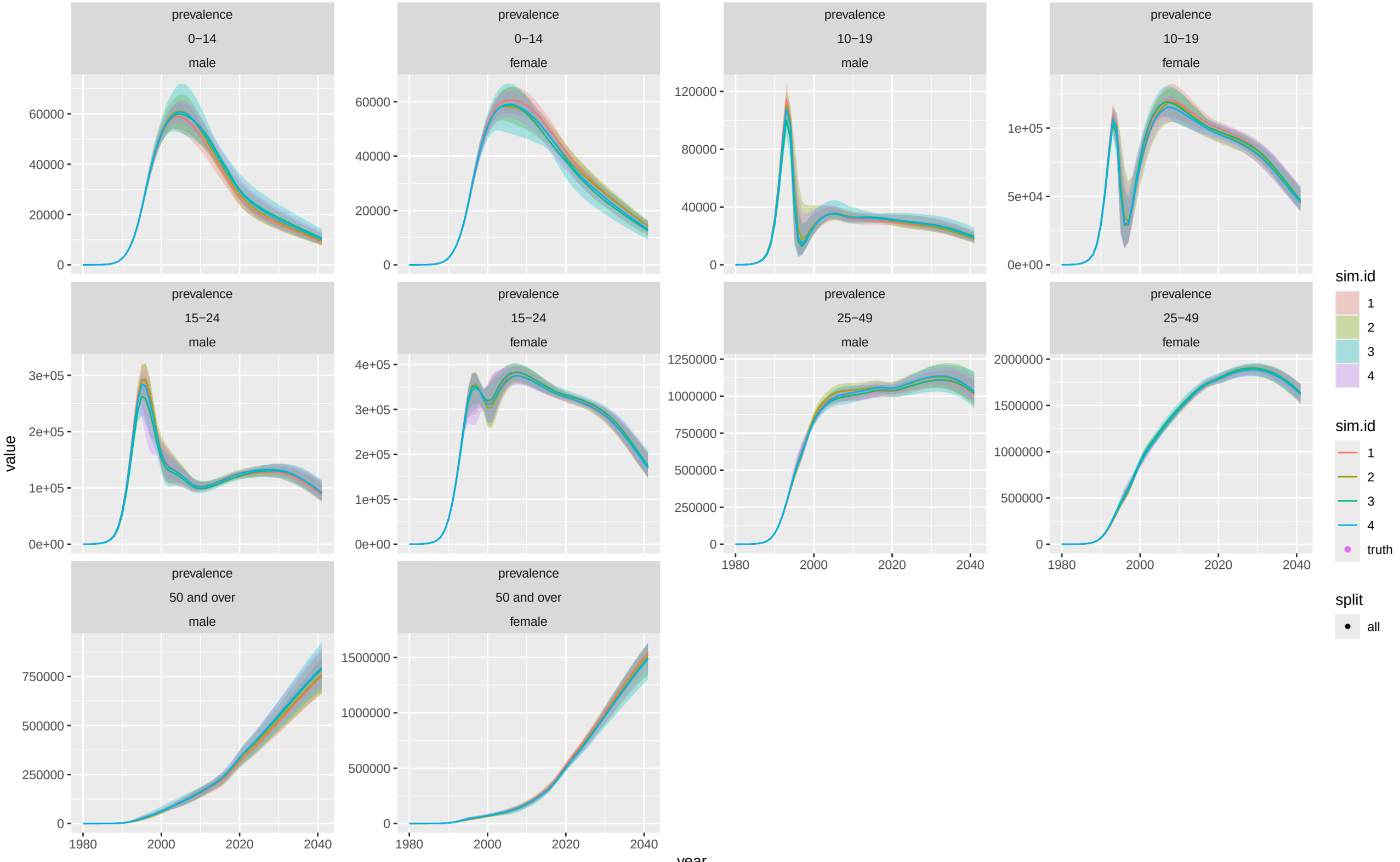
r1.upper.middle step 2: hiv.mortality.by.sex (one color per chain)



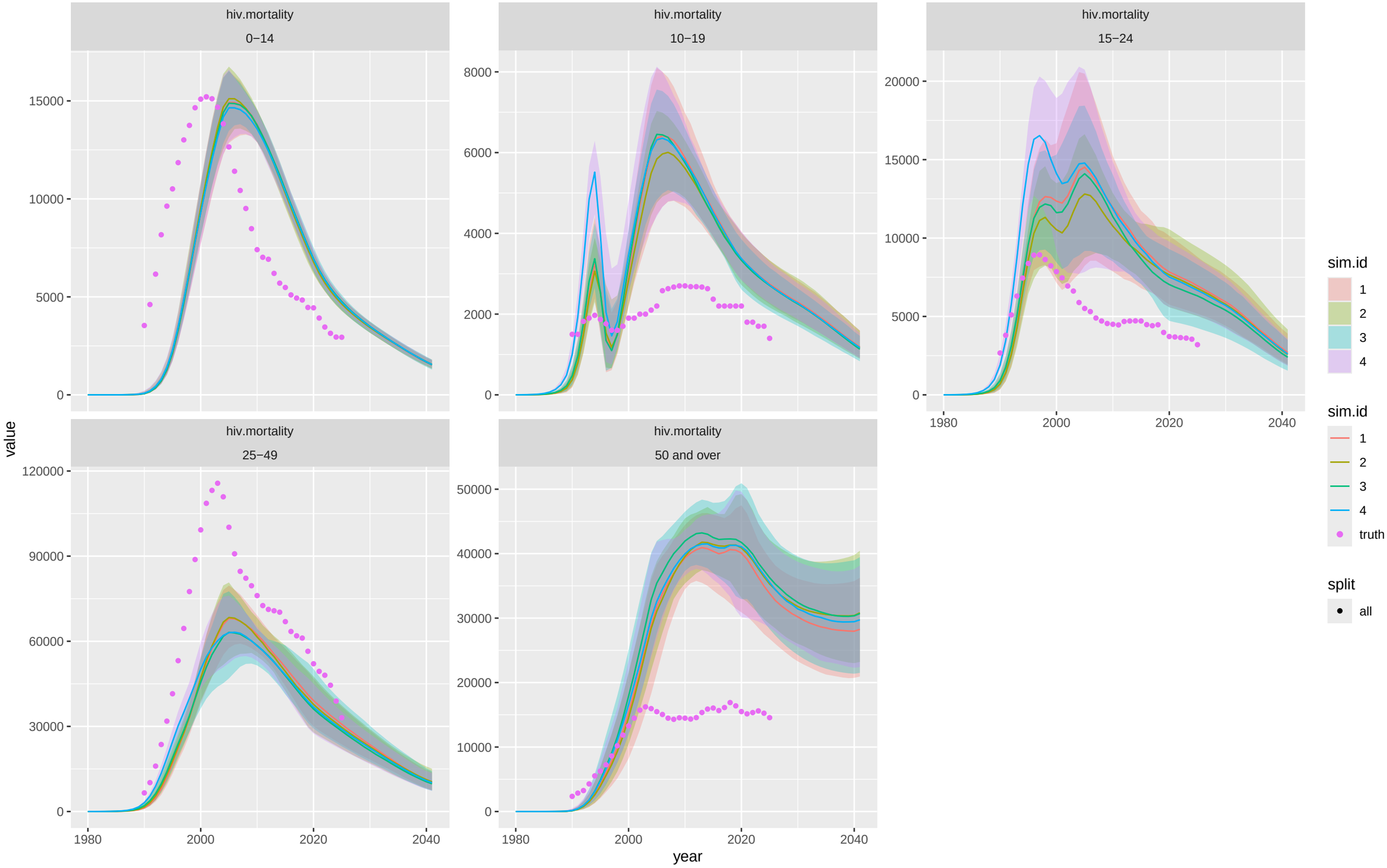
r1.upper.middle step 2: incidence.by.age.sex (one color per chain)



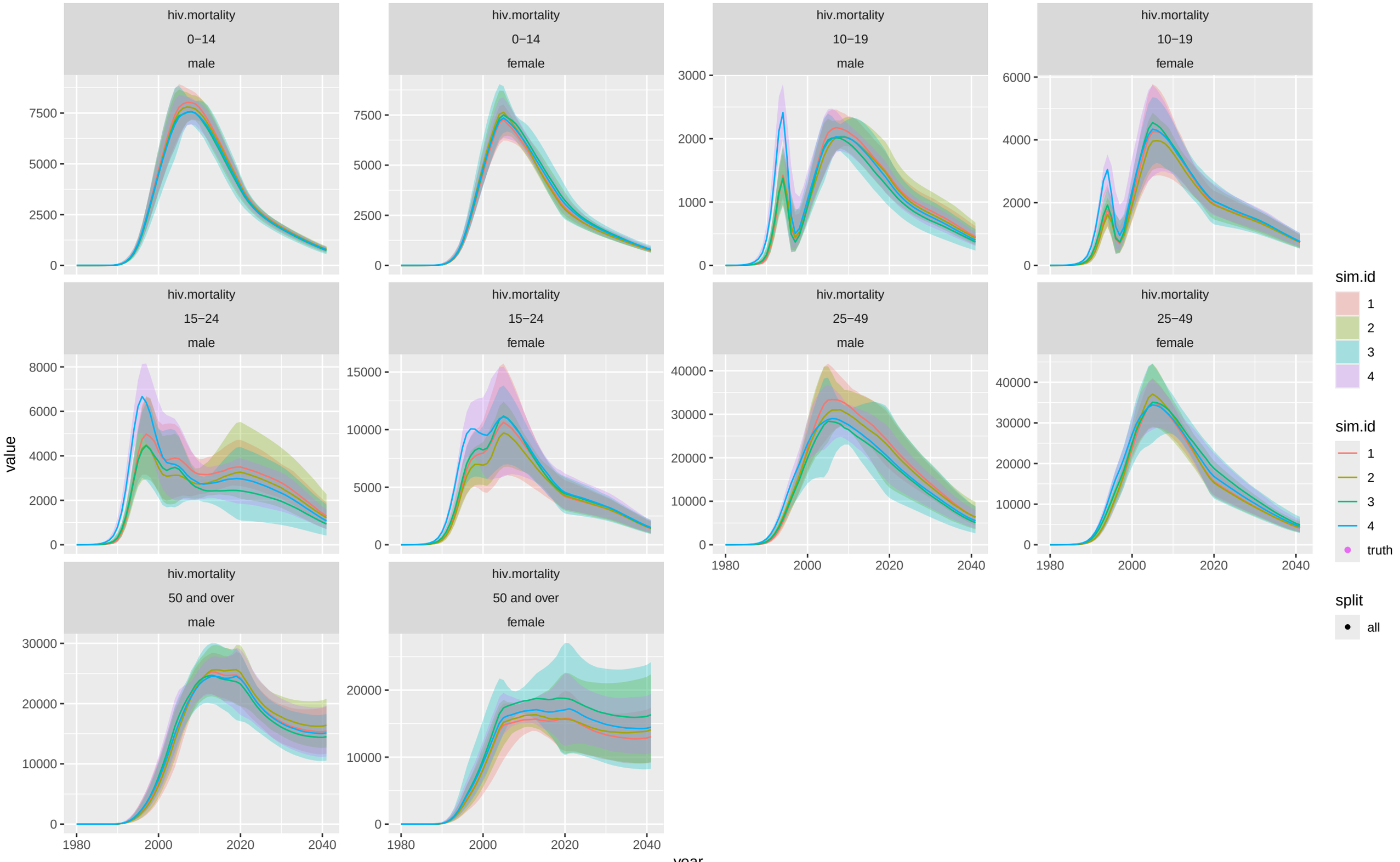
r1.upper.middle step 2: prevalence.by.age.sex (one color per chain)



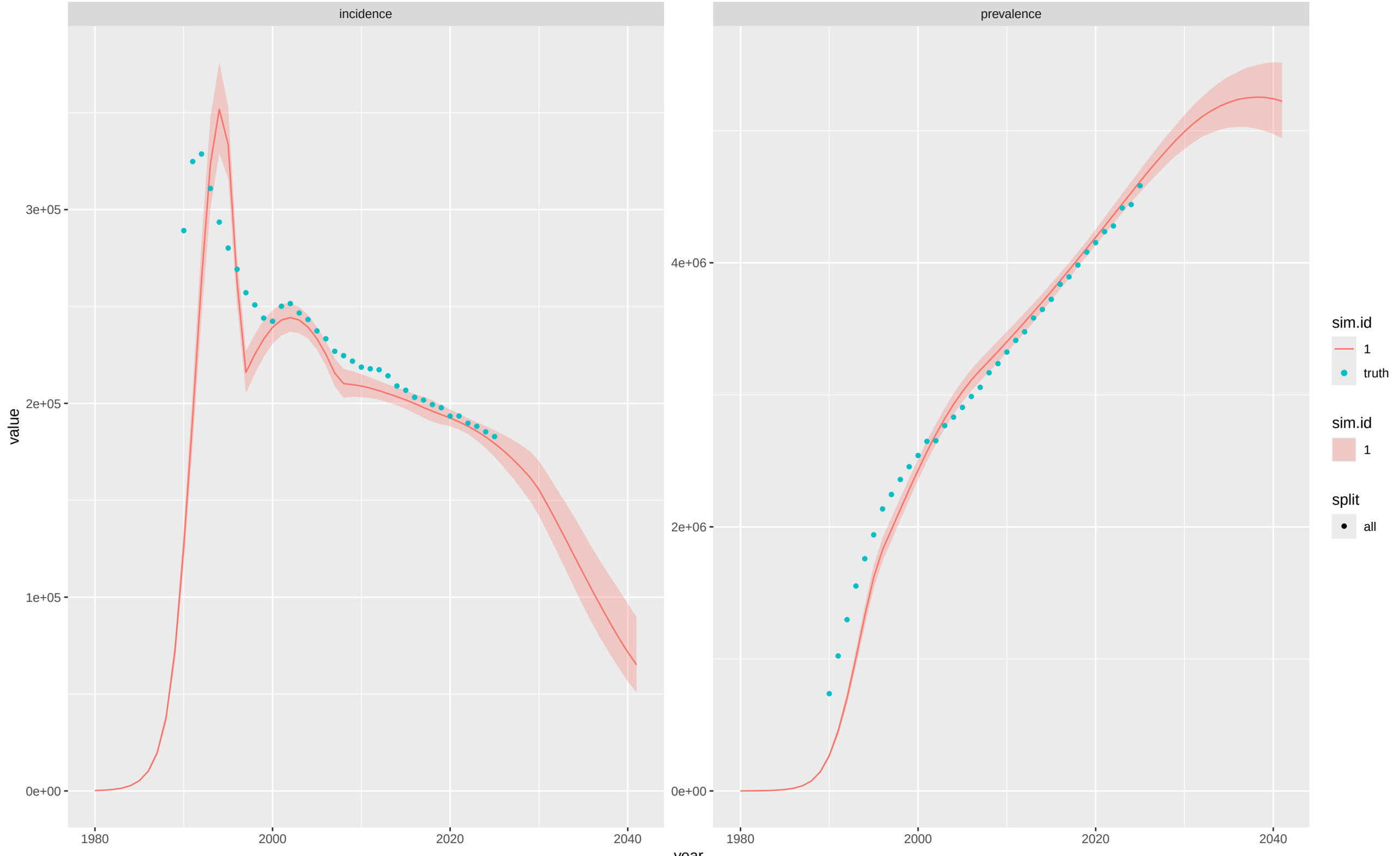
r1.upper.middle step 2: hiv.mortality.by.age (one color per chain)



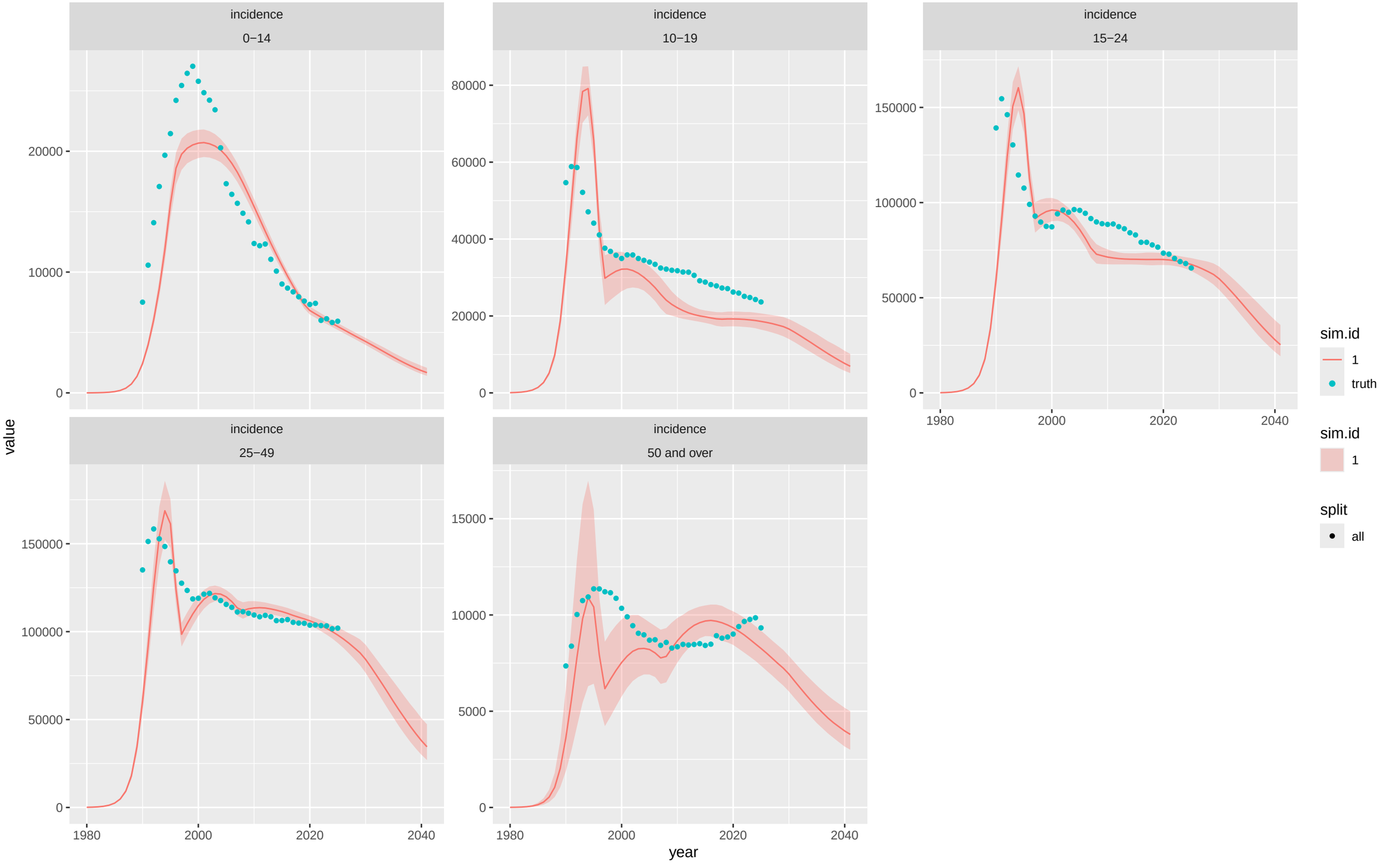
r1.upper.middle step 2: hiv.mortality.by.age.sex (one color per chain)



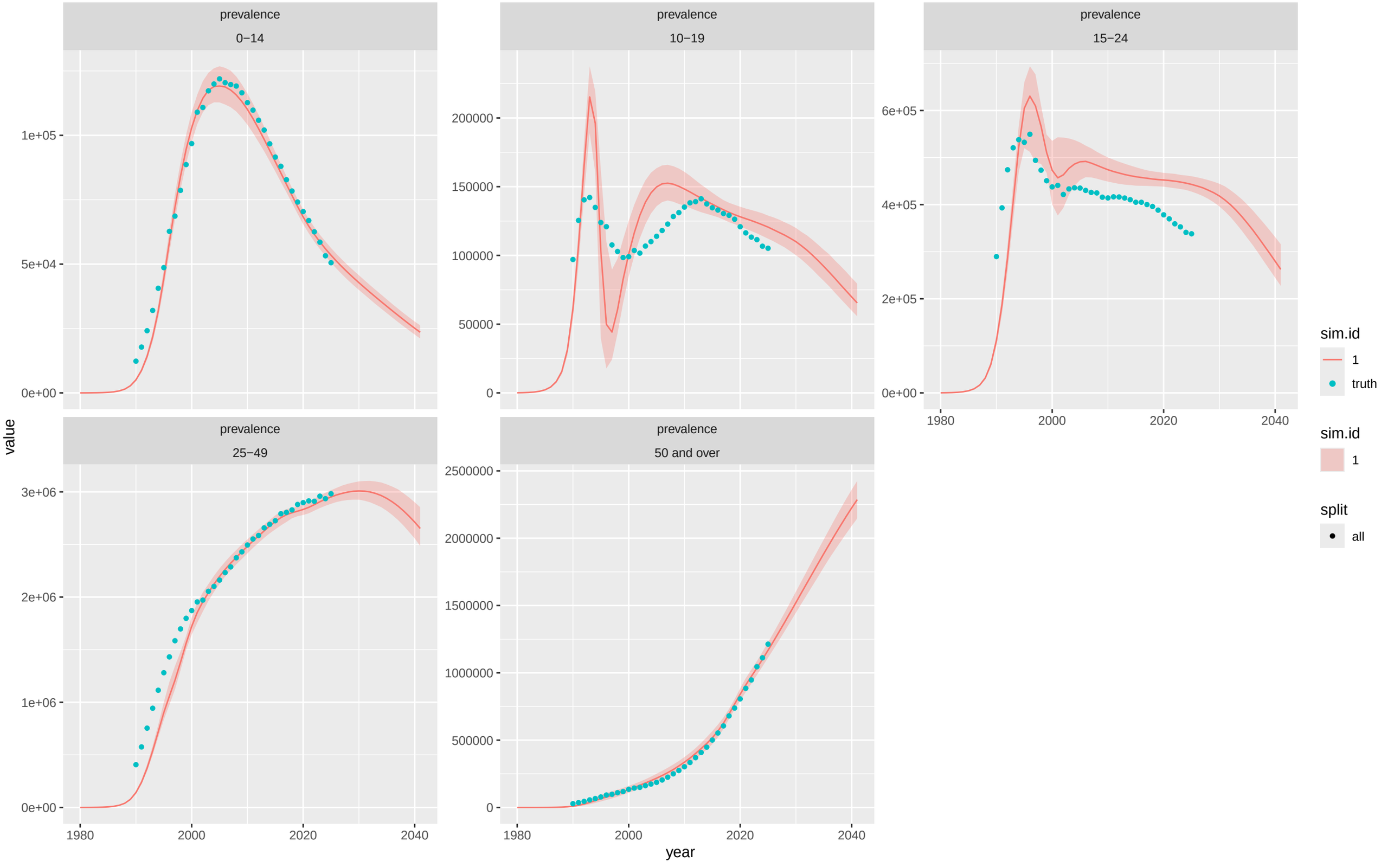
r1.upper.middle: fit (merged Step 2 simset)



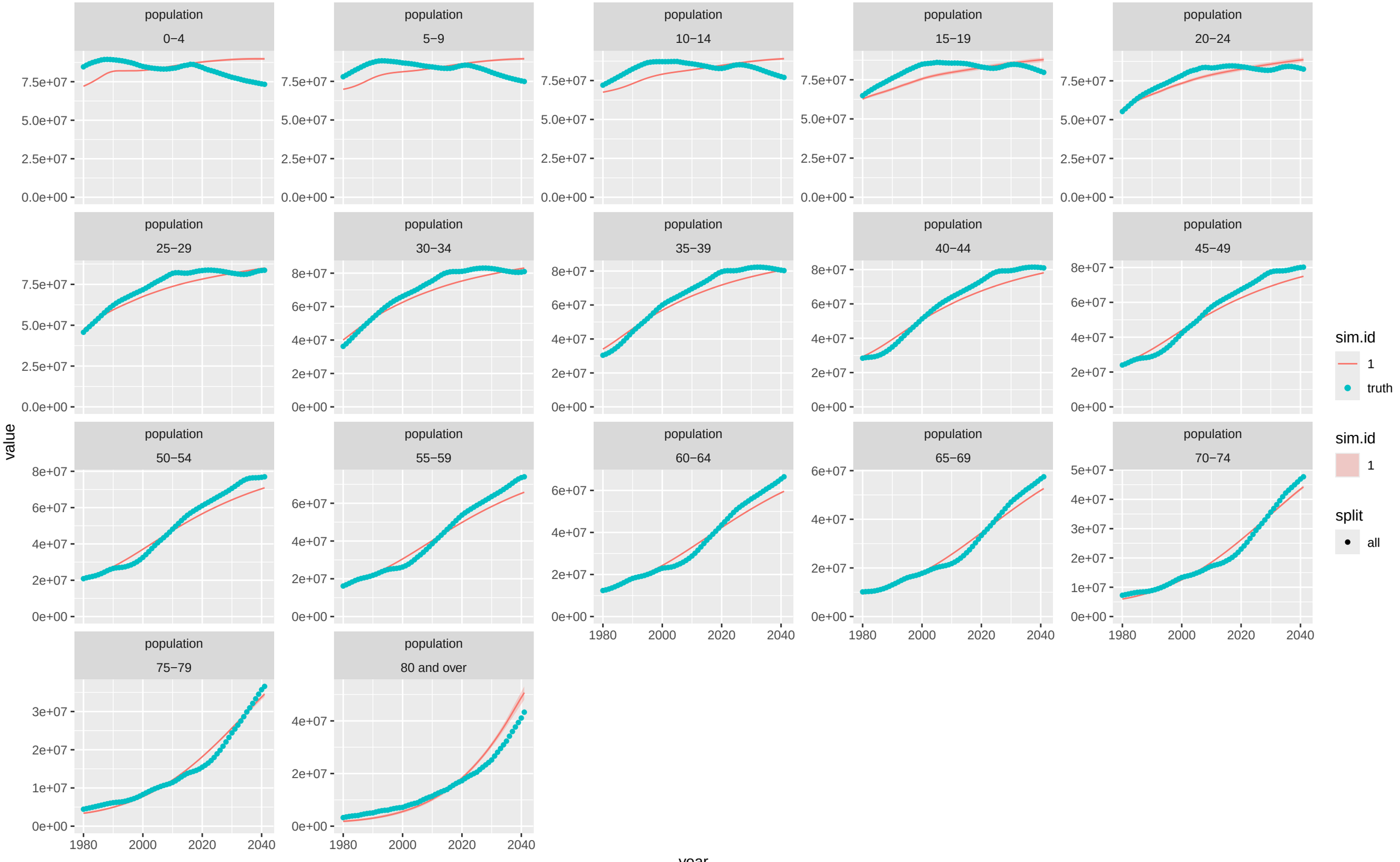
r1.upper.middle: incidence.by.age (merged Step 2 simset)



r1.upper.middle: prevalence.by.age (merged Step 2 simset)



r1.upper.middle: population.by.age (merged Step 2 simset)



r1.upper.middle: cascade (merged Step 2 simset)

awareness

engagement

suppression

value

year

1.00
0.75
0.50
0.25
0.00

1980
2000
2020
2040

sim.id
1
truth

sim.id
1

split
all

r1.upper.middle: cascade.by.age.sex (merged Step 2 simset)

awareness 0−14 male
awareness 0−14 female
awareness 15+ male
awareness 15+ female
awareness 15−24 male
awareness 15−24 female
engagement 0−14 male
engagement 0−14 female
engagement 15+ male
engagement 15+ female
engagement 15−24 male
engagement 15−24 female
suppression 0−14 male
suppression 0−14 female
suppression 15+ male
suppression 15+ female
suppression 15−24 male
suppression 15−24 female

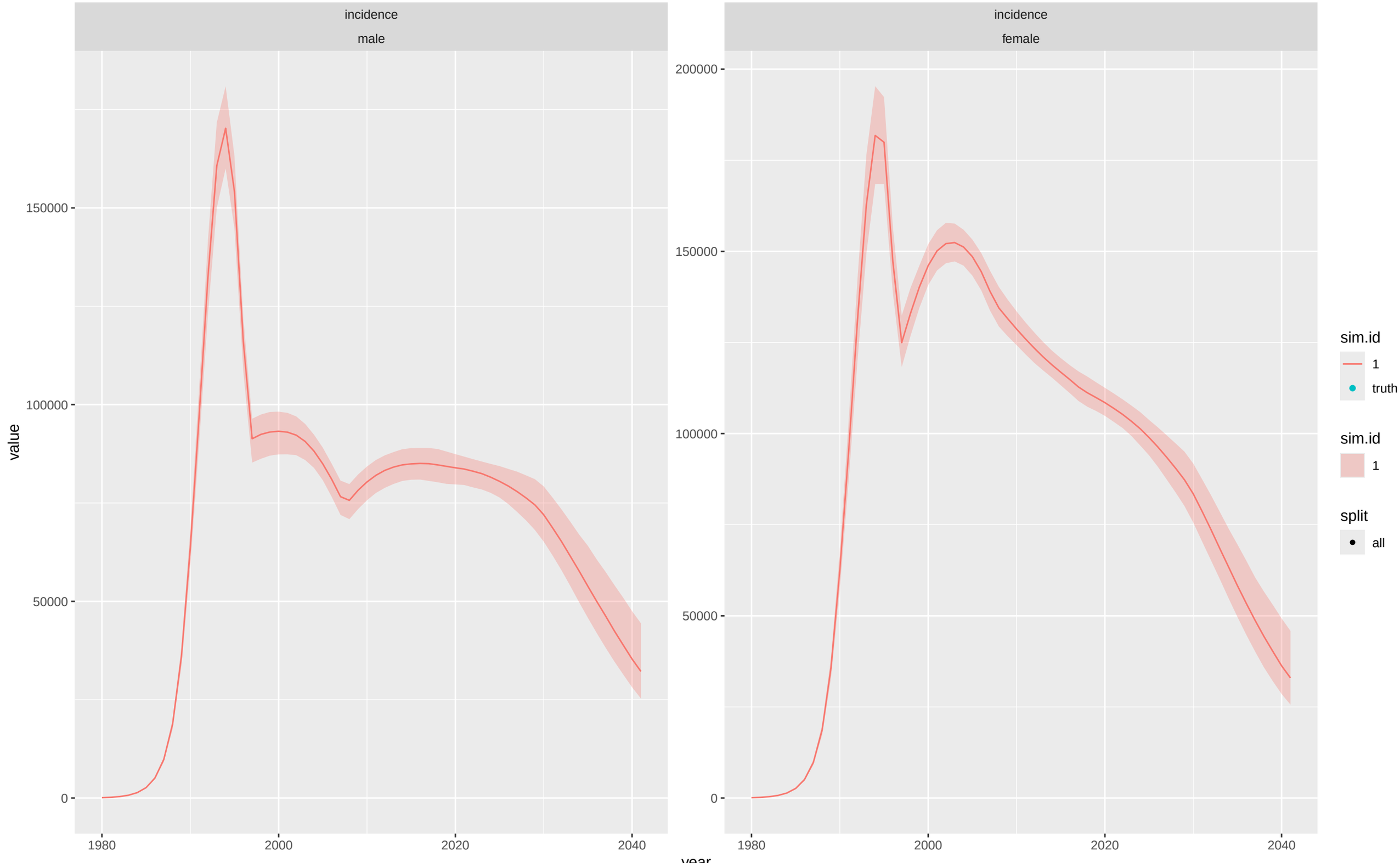
value

year

sim.id
1
truth

split
all

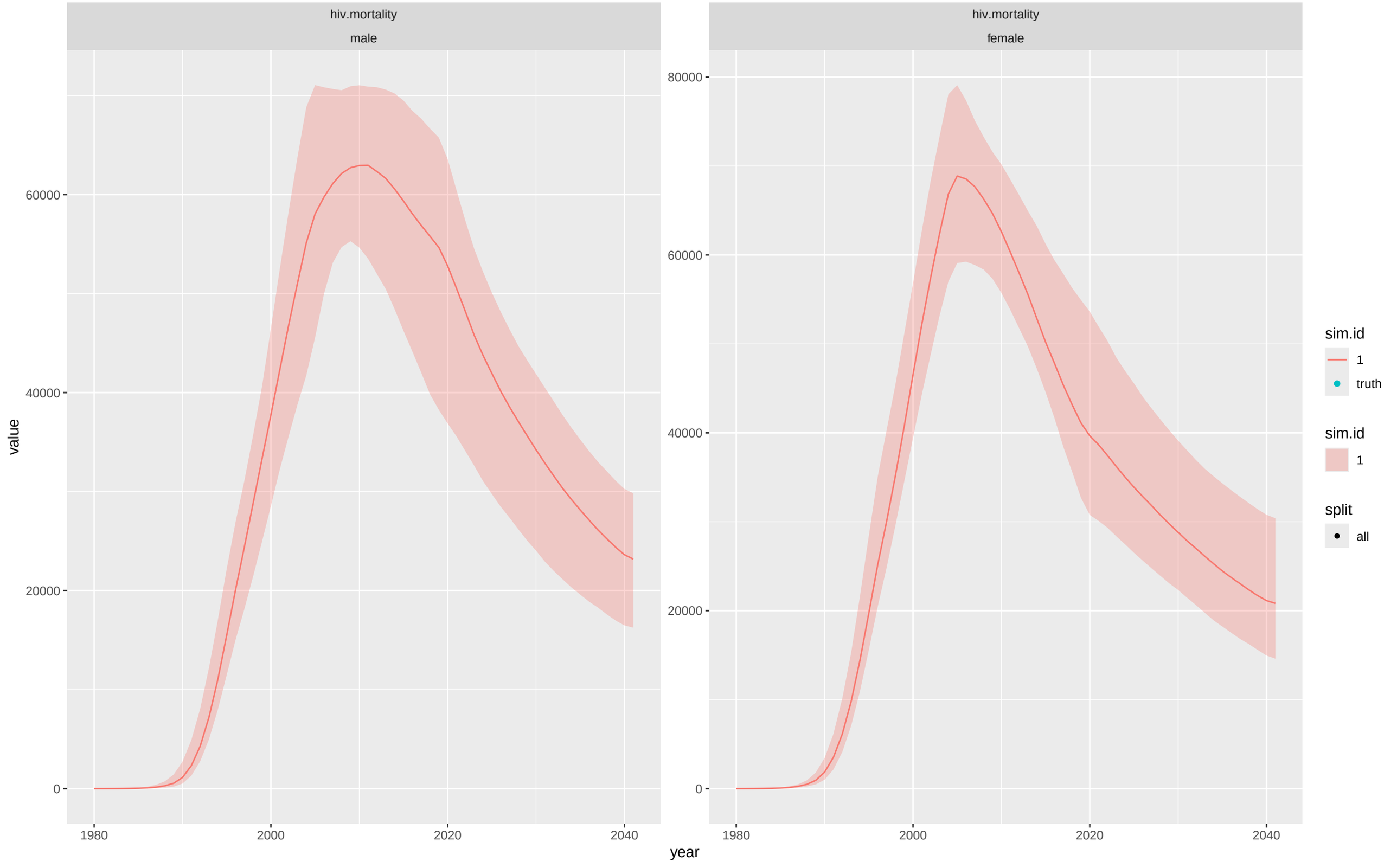
r1.upper.middle: incidence.by.sex (merged Step 2 simset)



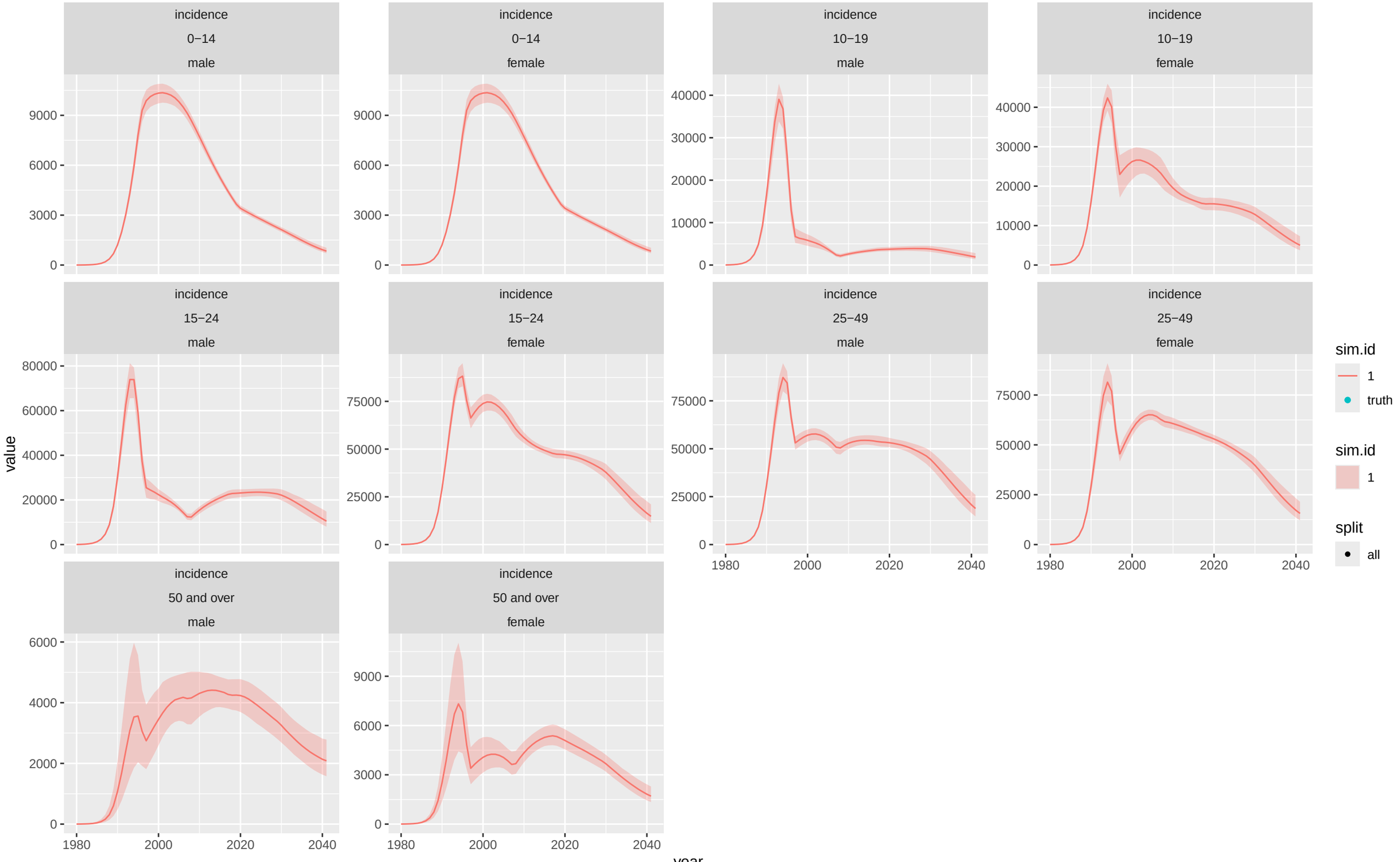
r1.upper.middle: prevalence.by.sex (merged Step 2 simset)



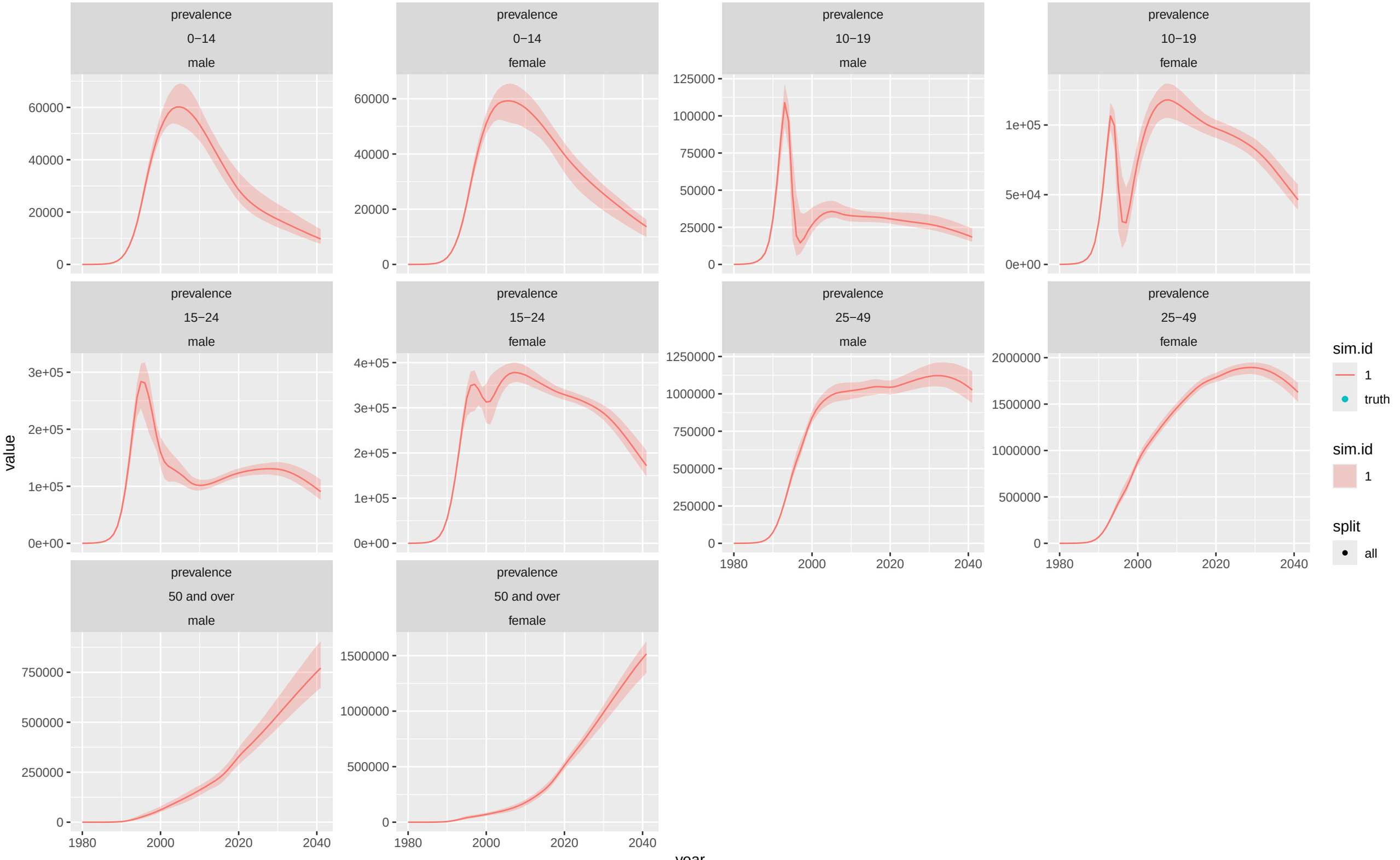
r1.upper.middle: hiv.mortality.by.sex (merged Step 2 simset)



r1.upper.middle: incidence.by.age.sex (merged Step 2 simset)



r1.upper.middle: prevalence.by.age.sex (merged Step 2 simset)



r1.upper.middle: hiv.mortality.by.age (merged Step 2 simset)



r1.upper.middle: hiv.mortality.by.age.sex (merged Step 2 simset)

hiv.mortality 0–14 male

hiv.mortality 0–14 female

hiv.mortality 10–19 male

hiv.mortality 10–19 female

hiv.mortality 15–24 male

hiv.mortality 15–24 female

hiv.mortality 25–49 male

hiv.mortality 25–49 female

hiv.mortality 50 and over male

hiv.mortality 50 and over female

value

year

sim.id
1
truth

sim.id
1

split
all